6.S974/2.S988/1.S980: Games, Learning, and Security

Wargame Scope

Overview

A typical wargame should contain:

- One well-defined operational setting.
- Multiple (2-4) strategically meaningful agent types.
- A dynamic environment evolving over approximately 20–100 decision periods.
- A manageable state space, often a network with roughly 10–100 nodes or assets.
- Explicit resource constraints and competing objectives.
- Partial or asymmetric information.
- At least one adaptive or learning agent.
- Two or three plausible attack or disruption mechanisms.
- Two or three defense or resilience interventions.
- Repeated simulation across scenarios and random seeds.

The project should use at least one credible method within each course module and integrate them coherently:

Course module	**Minimum expected implementation**
Strategic modeling	A game among at least two decision-makers, with explicit actions, information, and payoffs
Learning and adaptation	Some adaptive policies compared with a fixed or equilibrium-based policy
Attack and defense	An attacker who chooses targets, timing, or attack type, and a defender who allocates limited resources
Resilient design	Comparison of at least two system designs or operating policies under nominal and contested conditions

By the end of the semester, a user should be able to run the testbed and answer questions such as:

- How does strategic behavior change system outcomes relative to nonstrategic or random behavior?
- Does an equilibrium-based strategy outperform simple heuristics?
- What does a learning agent discover through repeated interaction?
- Does learning converge, cycle, collude, or exploit an unintended feature of the environment?
- Which assets, links, information channels, or policies create the greatest vulnerability?
- How does the attacker adapt after observing the defender?

- Which defense works against the modeled attacker, and does it fail against a different attacker?
- How much nominal efficiency should be sacrificed to improve resilience?
- Does the recommended design remain effective under rare events, model misspecification, and previously unseen attack strategies?

Four Suitable Example Projects

Example 1: Contested Logistics Operations

A defender routes limited vehicles to deliver supplies across a network. An adversary can observe some movements and disrupt a limited number of links or facilities.

The final wargame could:

- compare shortest-path, robust-optimization, and adaptive routing policies;
- model a strategic attacker using network interdiction;
- allow the attacker to learn the defender's recurring patterns;
- test deception or route randomization;
- redesign the network through prepositioning, redundancy, or flexible transportation capacity;
- quantify fulfilled demand, delay, cost, and worst-case mission failure.

A strong final conclusion might identify when randomized routing is worth its additional cost and whether prepositioning is more valuable than transportation redundancy.

Example 2: Autonomous Maritime Surveillance

Several autonomous vehicles patrol a region while an intruder attempts to cross undetected. Vehicles have limited energy, imperfect sensors, and uncertain communication.

The final wargame could:

- model the interaction as a security or pursuit–evasion game;
- compare fixed patrols, equilibrium patrols, and learned patrol policies;
- introduce sensor spoofing, communication loss, and deceptive intruder behavior;
- examine whether predictable learning creates exploitable patrol patterns;
- test human oversight, randomized patrols, and redundant sensing;
- evaluate detection probability, energy consumption, false alarms, and recovery after failure.

Example 3: Strategic Microgrid Defense

A microgrid operator allocates generation, storage, and reserve capacity while an attacker disrupts components and uncertain demand evolves.

The final wargame could:

- model strategic attacks on generators, transmission links, or sensors;
- compare deterministic dispatch, robust dispatch, and adaptive control;
- model cascading failures following an initial disruption;
- allow the attacker to infer the operator's reserve-allocation policy;
- test storage, islanding, redundancy, and load-shedding priorities;
- measure operating cost, unmet demand, recovery time, and tail risk.

The final study might show that the least-cost nominal design is structurally fragile and determine the minimum resilience investment needed to control catastrophic loss.

Example 4: Algorithmic Market with Adaptive Agents

Multiple pricing or bidding agents repeatedly compete in a stylized market. A regulator attempts to preserve competition and welfare.

The final wargame could:

- compare myopic, no-regret, and reinforcement-learning agents;
- examine convergence, cycling, or tacit collusion;
- introduce strategic manipulation or information asymmetry;
- test monitoring, penalties, randomized audits, or alternative market rules;
- evaluate prices, profits, consumer welfare, volatility, and fairness;
- investigate whether an intervention effective against one learning algorithm generalizes to others.

Suggested Final Deliverables

Each completed wargame should include the following.

1. Reproducible Computational Testbed

A documented Python implementation that allows the instructors to:

- modify important scenario parameters;
- select strategies for the agents;
- run a single game or a batch of replications;
- save performance metrics and agent actions;
- reproduce the principal figures and tables.

A graphical interface can be encouraged, but may not necessarily be required. A clean command-line interface or notebook is sufficient.

2. Explicit Game Specification

Students should clearly define:

- agents and their objectives;
- system state;
- available actions;
- sequence and timing of decisions;
- information available to each agent;
- uncertainty and transition dynamics;
- payoff and performance functions;
- termination conditions.

This should be treated as a primary intellectual deliverable, not merely implementation documentation.

3. Several Policy Classes

At minimum:

- a simple benchmark, such as random, greedy, or rule-based behavior;
- a strategic or optimization-based policy;
- some adaptive or learning policies for at least one agent.

4. Attack–Defense Experiments

The wargame should include:

- a nonstrategic disruption baseline;
- a strategic attacker;
- at least one adaptive or previously unseen attacker;
- multiple defense policies.

5. Resilient Redesign

Students should change something more fundamental than the operating policy, such as:

- network topology;
- capacity allocation;
- redundancy;
- sensing or information availability;
- recovery capability;
- incentives;

- human-oversight rules.

They should explicitly measure the efficiency–security–resilience tradeoff.

6. Stress Testing

The final study should include:

- nominal conditions;
- strategic attacks;
- stochastic disruptions;
- rare or high-impact scenarios;
- sensitivity analysis;
- out-of-distribution strategies or parameters.

A reasonable target would be 20–50 experimental configurations, each evaluated over enough replications to report means, variability, and tail outcomes.